pfi

Python for Fixed Income

COURSE 4 · MACHINE LEARNING

The ML Workflow and Simple Regression

WEEK 1 · DAY 1

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The question

How much of a bond's spread can one characteristic account for?

821

corporate bonds

150

issuers, all invented

AAA to CCC

the ratings

OAS (bp)

the target

Rating score

1 for AAA up to 18 for CCC, one step per notch

Duration

Modified duration, in years

Coupon

Annual coupon, in % of par

The answer has to hold on bonds the model has never seen.

First look: complete and clean

```
data.head()
data.shape
(821, 21)
data.info()
data.describe().T
data.isnull().sum()
print(data["cusip"].duplicated().sum())
0
```

821 x 21

bonds by columns

0

missing values

0

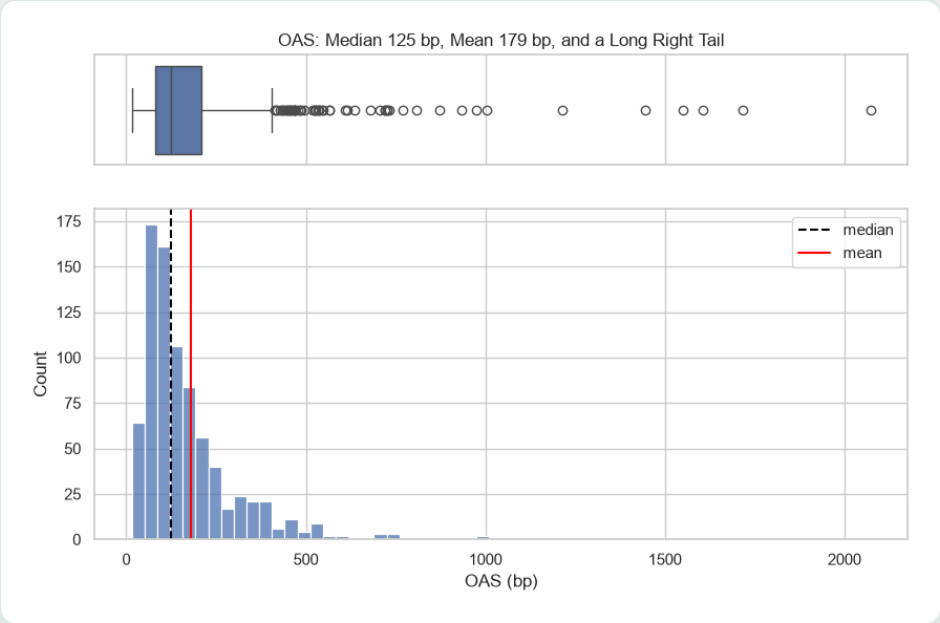
CUSIPs in the file twice

125 / 179

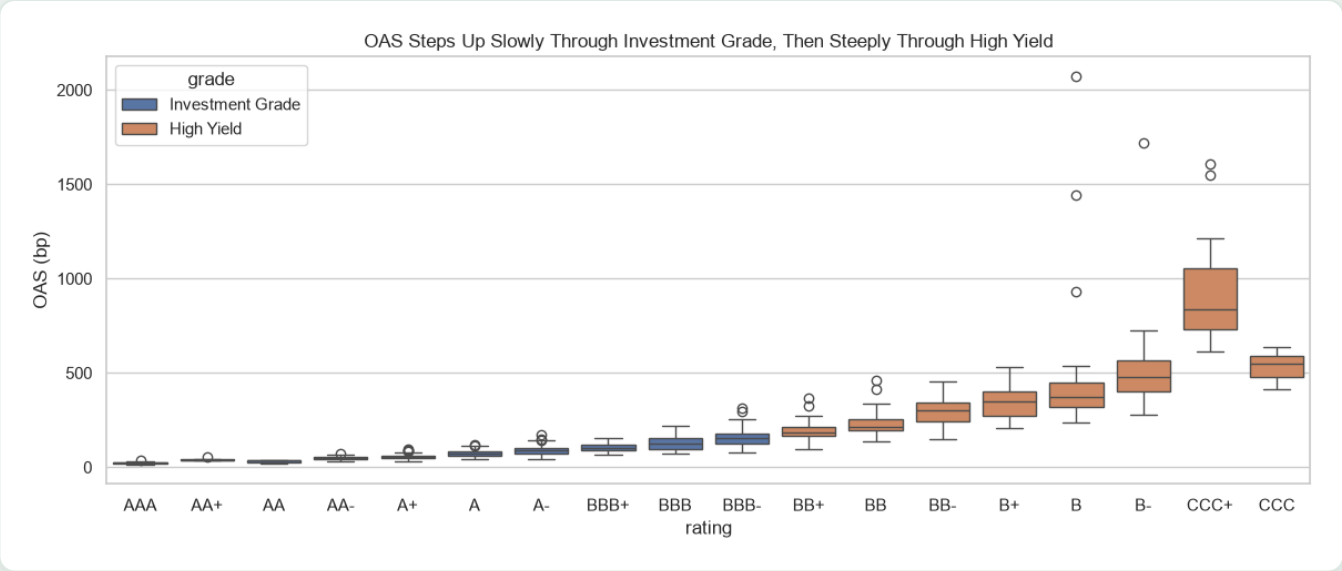
median and mean OAS, bp

Nothing to fix, so we go straight to the charts.

A long tail, and a curve by rating



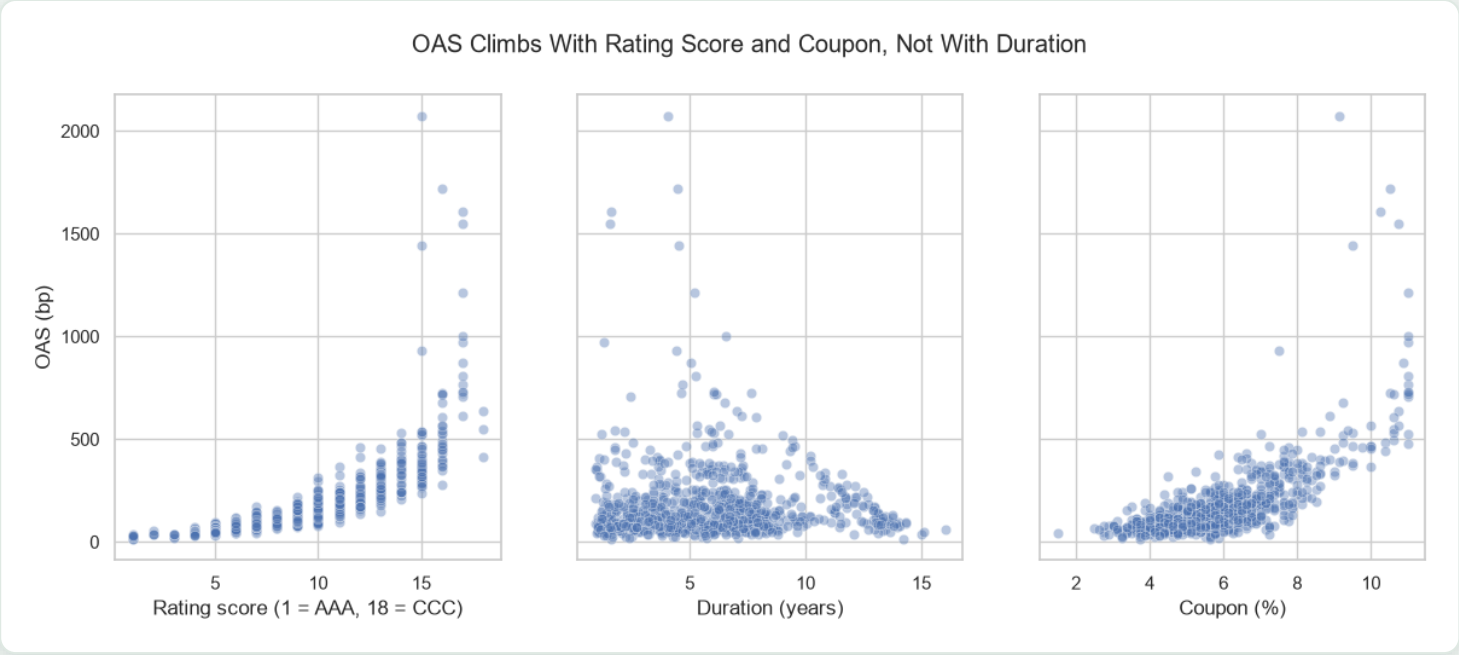
Middle half: 79 to 210 bp. Above 1,000 bp: 7 bonds.



Small steps through investment grade, big ones through high yield. Thin ends: CCC 3 bonds, AA+ 4, AA 5.

A straight line will struggle at both ends. We fit one anyway and measure the miss.

Three candidates



Correlation with OAS

Column	With OAS
Rating score	0.73
Coupon	0.72
Duration	-0.06
yield_pct	0.99
dts	0.79

Yield and DTS are built from OAS itself, so they stay out.

Spread climbs with rating score and coupon. Against duration the cloud is flat.

Lock away a quarter of the bonds

```
X = data[["score", "duration", "coupon"]]
y = data["oas"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=SEED)
Training bonds: 615  Test bonds: 206
```

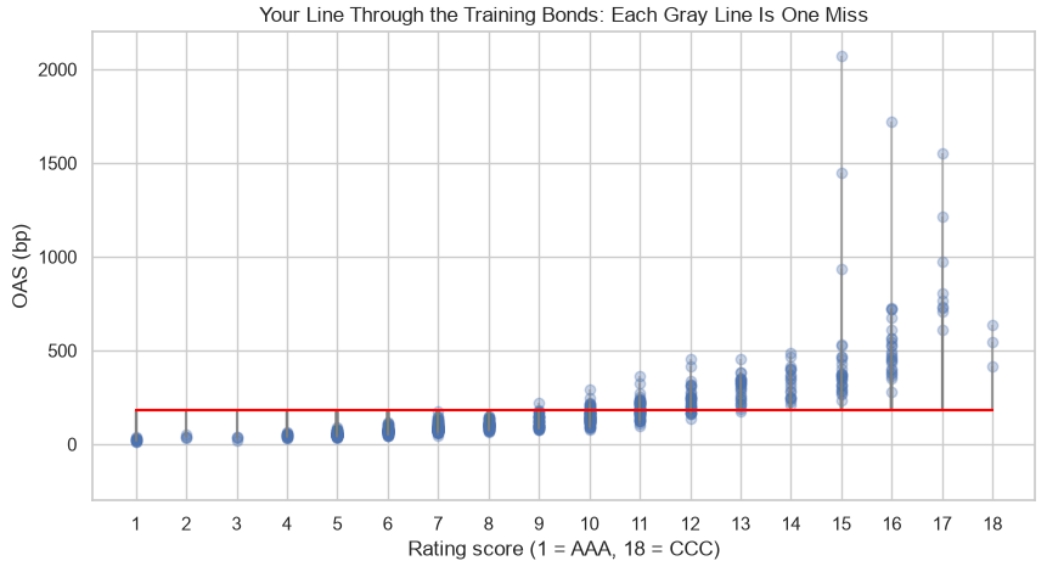
615 training bonds: every model is fitted here

206 test bonds

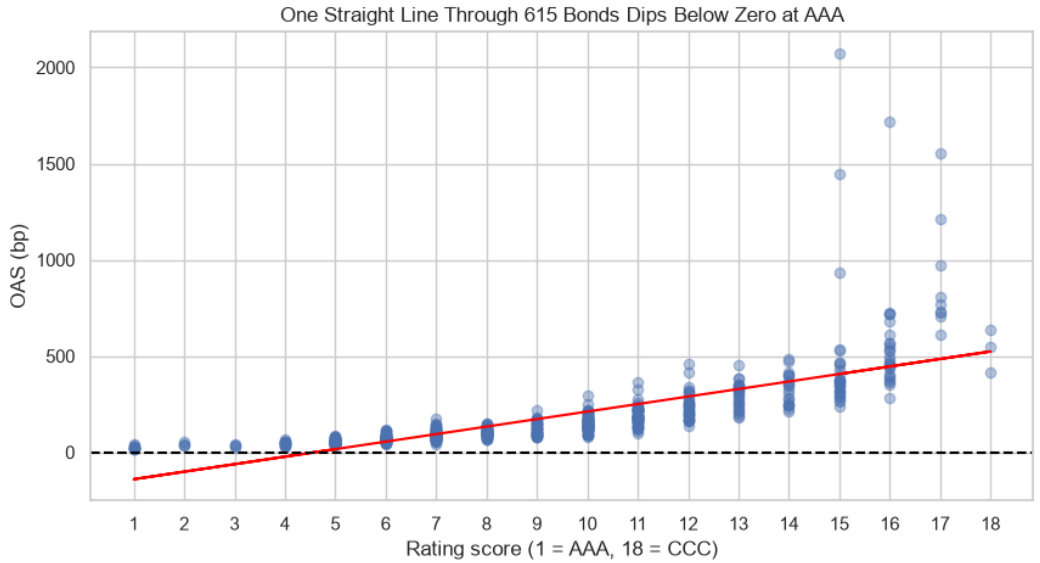
The test bonds stay untouched until the models are scored: that score is the only honest one.

Your turn, then the computer's

1. By eye: the flat start misses 115.1 bp



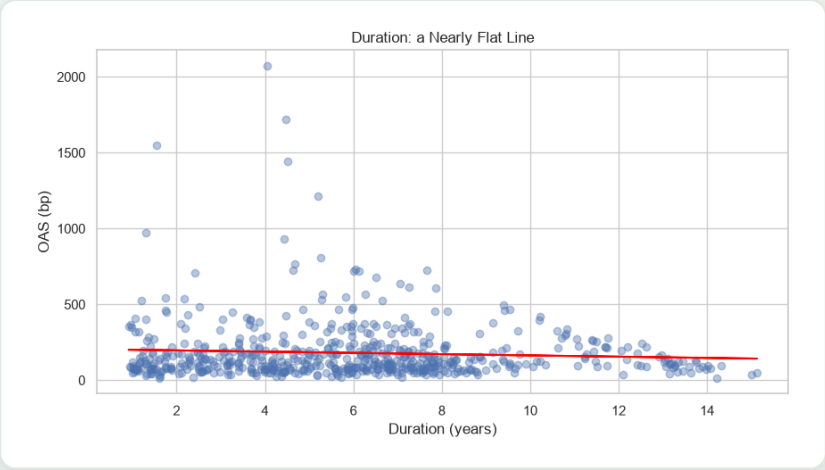
2. Least squares turns the same two knobs.

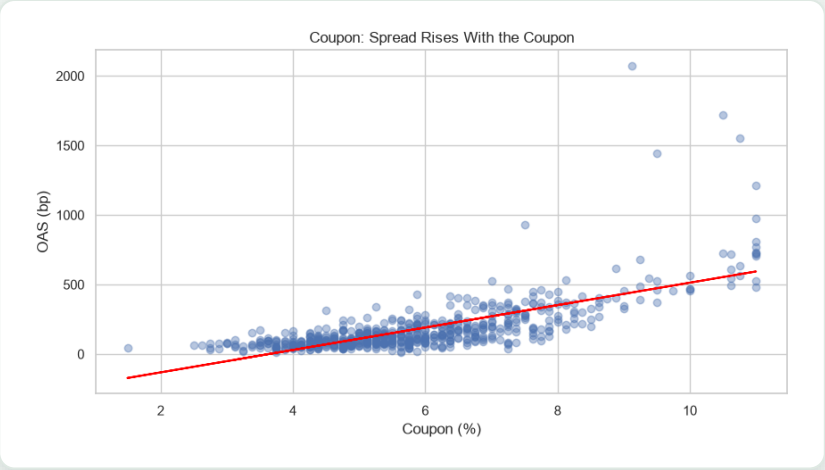



```
model_score = LinearRegression()
model_score.fit(X_train[features_score], y_train)
OAS = ( 39.05 ) * score + ( -179.06 )
```

One line per candidate

Feature	The fitted line	Training R-squared
Rating score	`OAS = ( 39.05 ) * score + ( -179.06 )`	0.523
Duration	`OAS = ( -4.13 ) * duration + ( 205.11 )`	0.005
Coupon	`OAS = ( 80.62 ) * coupon + ( -292.77 )`	0.504





Rating: 39.05 bp more fitted spread per notch. Duration alone: next to nothing.

Scored on bonds they have never seen

Five measures, one function

Measure	What it reads
MAE	the average miss, bp
RMSE	long misses count more, bp
MAPE	miss as % of the spread
R-squared	share of variation

The 206 test bonds

Line	MAE (bp)	RMSE (bp)	R-squared
Rating score	57.7	108.3	0.553
Duration	97.8	162.3	-0.003
Coupon	64.0	107.9	0.556

```
score_test_perf = model_performance_regression(model_score, X_test[features_score], y_test)
```

Rating score has the smallest typical miss, and its test MAE is close to its training MAE (65.6 bp): no overfitting.

The memoriser

Seen bond: it reads its notes.

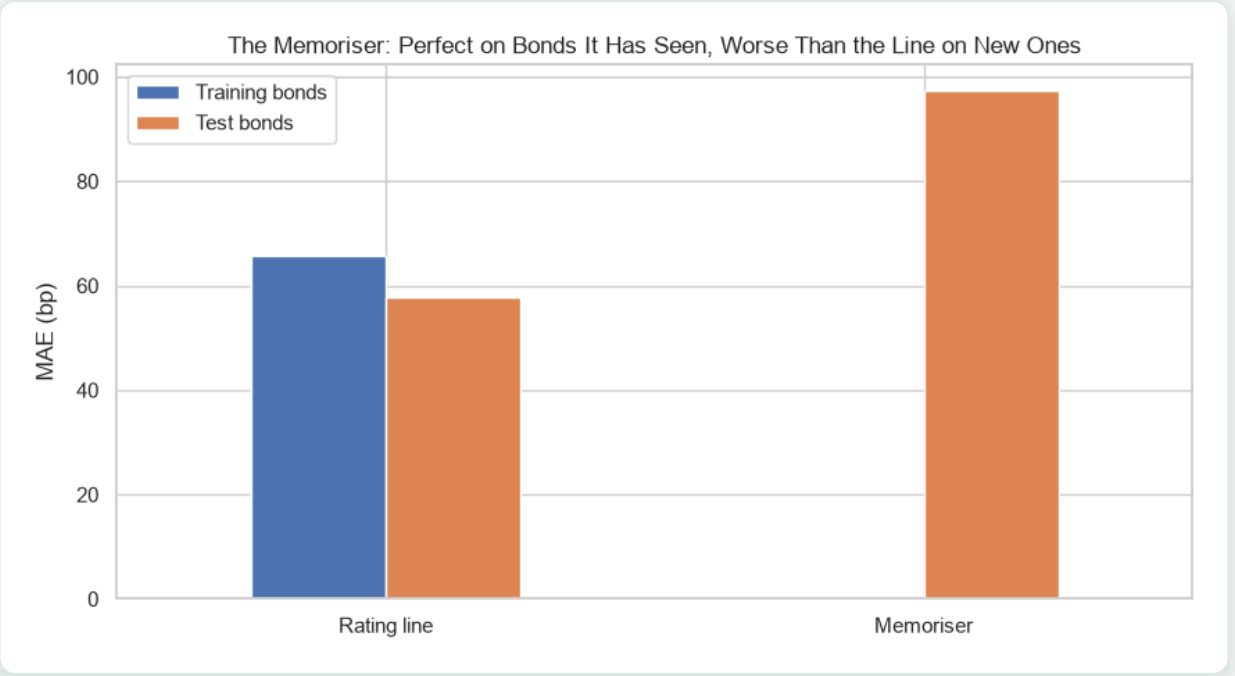
New bond: it says the training average.

0.0 bp

MAE on bonds it has seen

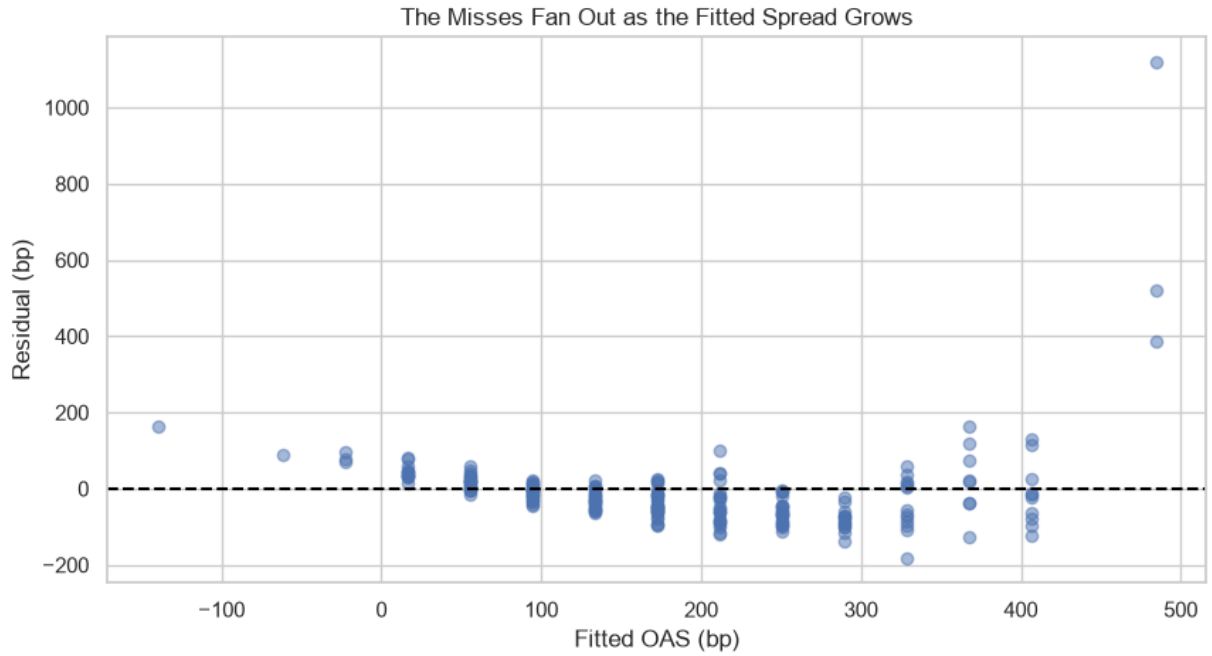
97.4 bp

MAE on new bonds



A perfect training score proves nothing. A model is only as good as its score on bonds it has never seen.

Where the line misses



The largest miss

Praxiskan Finance, CCC+

Actual 1,605 bp, fitted 484.8 bp

Miss: 1,120.2 bp

52 percent of the squared test miss

A U shape: too low at both ends, too high in the middle. The misses grow with the spread.

A residual is the model's miss on one bond, not a view on the bond.

What the three lines say

39.05 bp

more fitted spread per notch

57.7 bp

the rating line's typical miss on unseen bonds

52%

of the training variation, from rating alone

203 of 206

test bonds share an issuer with a training bond

Coupon comes close: it carries part of the spread itself.

Duration alone says almost nothing.

The rating line is straight on a curve: below zero at AAA, short of the widest bonds.

One straight line on rating carries about half the story.

What you can do now

- Run a data overview and read the charts before fitting
- Fit a simple linear regression and read its equation
- Score each model in bp on bonds it has never seen
- Tell a model that learned from one that memorised
- Read a residual plot for the shape a line misses

Day 2 Multiple Regression and Dummy Variables: every rating gets its own step, and duration joins the rating.